

App Development Project Report

MAD II Project Report

1. Student Details

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About Me: I am currently pursuing the BS Degree program at IIT Madras and have a strong interest in web application development and data-driven technologies. I enjoy creating impactful applications that integrate learning, data analysis, and an engaging user experience.

2. Project Details

Project Title: Placement Portal Application (PPA)

Problem Statement:

To design and develop a web-based that enables institutes to efficiently manage campus recruitment activities, including company registrations, placement drives, student applications, and recruitment tracking through a centralized platform for Admins, Companies, and Students.

Approach:

The app was built using Flask as the backend framework with a modular structure along with Vue.js as the frontend framework. It allows institutes to efficiently manage campus recruitment activities including company approvals, placement drive management, and student applications.

3. AI/LLM Declaration

I used **ChatGPT** and **Claude** to assist with writing some SQLAlchemy model definitions, generating API documentation examples, and UI/styling suggestions. The extent of AI/LLM usage was approximately **10–15%**, mainly limited to **code suggestions** and **documentation formatting**. All core implementation logic, debugging, testing, and system integration were performed manually.

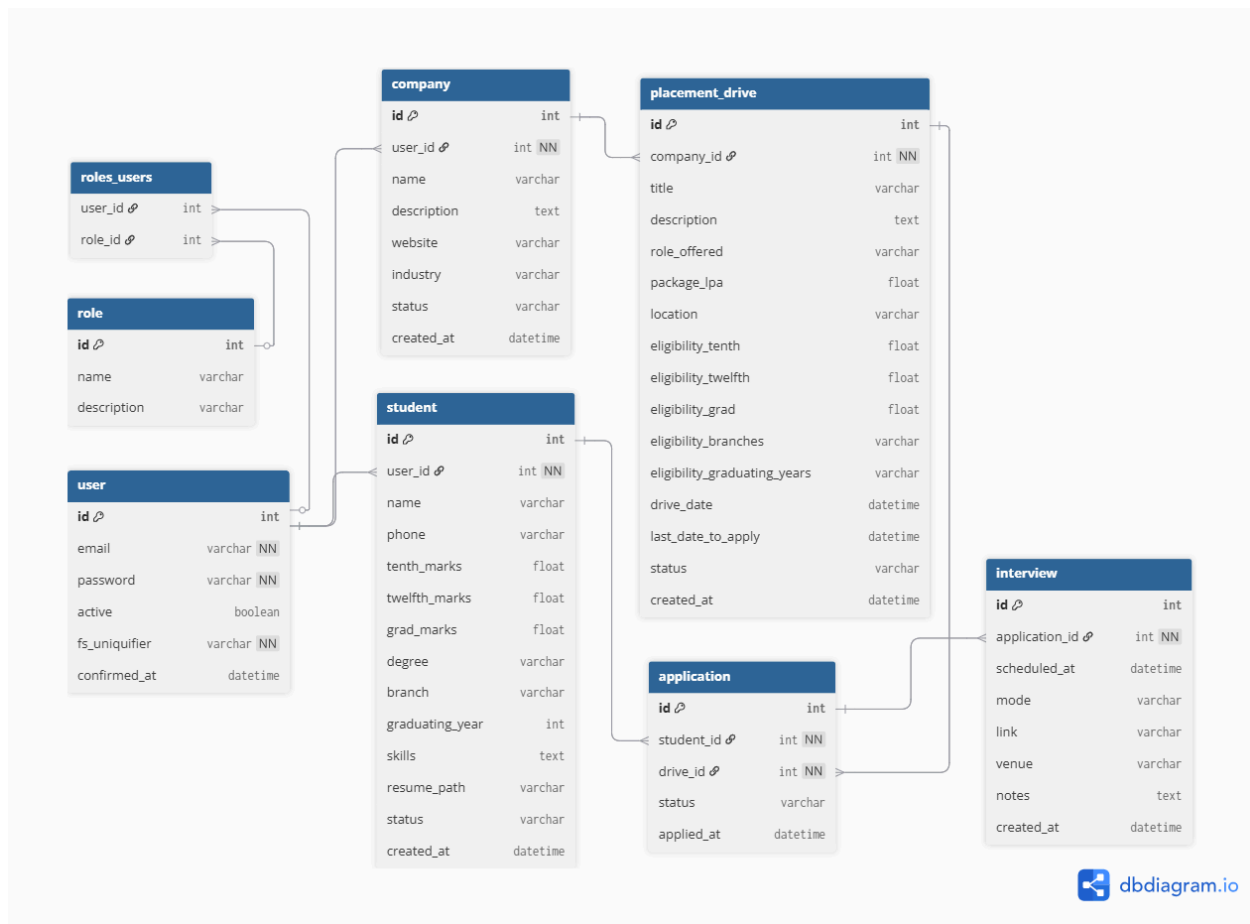
4. Technologies and Frameworks Used

Technology / Library	Purpose
Flask	Core backend web framework
SQLAlchemy	Object Relational Mapper for SQLite database
Vue.js	JS framework used to build the user interface
Bootstrap 5	Frontend styling and responsive design
Chart.js	To visualize trends such as activity vs. time
Flask-Login	User authentication and session management
Redis + Celery	For jobs (scheduled & async tasks)
SQLite	Lightweight local database for storing user data
Flask-Caching + Redis	For caching and performance

5. Database Schema / ER Diagram

Tables:

1. **User** — stores authentication and account details
2. **Role** — stores different user roles in the system such as Admin, Company, and Student
3. **Roles_Users** — association table used to map users with their assigned roles
4. **Company** — stores company profile details registered on the platform
5. **Student** — stores student profile and academic information
6. **PlacementDrive** — stores recruitment drive details created by companies
7. **Application** — records student applications to placement drives
8. **Interview** — stores interview scheduling and related details for shortlisted applications



6. API Resource Endpoints

Endpoint	Method	Description
<code>/api/auth</code>	POST, GET	Handles authentication functions such as user registration, login, logout, and retrieving the current user profile
<code>/api/admin</code>	GET, PUT, DELETE	Provides administrative operations including managing students, companies, placement drives, approvals, and reports

<code>/api/company</code>	GET, POST, PUT, DELETE	Allows companies to manage their profile, create and manage placement drives, view applicants, and update recruitment statuses
<code>/api/student</code>	GET, POST, PUT, DELETE	Enables students to manage their profile, browse drives, apply to drives, withdraw applications, and view placement history
<code>/api/student/ export-applications</code>	POST	Triggers an asynchronous background job to export the student's application history as a CSV file

7. Video Presentation

Drive Link:

<https://drive.google.com/file/d/14tknsHFCHK7x6c7WSECPQ2x6SnC-jGqG/view?usp=sharing>
